

Open Alignment Test Factory - cycles 10-12

2026-05-14

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Open Alignment Test Factory - cycles 10-12

Abstract

Cycles 10-12 were closure cycles. They did not add new task families, code, tests, figures, or benchmark artifacts. Instead, each cycle independently confirmed that the Open Alignment Test Factory campaign had already met its success criteria through milestone M-8 and that no further research, build, test, or audit work remained for this campaign.

The closure decision is consistent across the supplied researcher, worker, and auditor sessions. The validated result is a runnable, trace-centered alignment evaluation prototype for agentic systems: a high-level safety property is converted into a provider-agnostic TaskSpec, exercised through benign scripted trajectories or runtime traces, checked for trace integrity and task-specific trace requirements, scored by deterministic evidence-linked scorers, exported through Inspect-compatible logs and summaries, stress-tested for benchmark failure modes, and documented in developer-facing final reports [1].

The only workspace update made during this reporting cycle was a MANIFEST.md refresh so the artifact inventory includes the cycles 7-9 periodic report now present on disk. No report Markdown file was written by the reporter; this report is supplied for the orchestrator to write and render.

Introduction

The original mission was to determine whether an open-source developer could build a practical alignment test factory for agentic AI systems rather than another static jailbreak list. The core question was whether high-level safety properties could be turned into safe, reproducible, nontrivial evaluation tasks for agents that use tools, memory, delegation, uncertainty handling, and multi-step plans.

By the start of cycle 10, the campaign had already completed the implementation ladder:

- M-1 mapped the open-source alignment-evaluation landscape and identified agentic

gaps.

- M-2 defined an operational failure taxonomy and benchmark-quality rubric.
- M-3 created a provider-agnostic task specification schema and safe examples.
- M-4 built a deterministic toy runtime and trace capture path.
- M-5 connected the prototype to an Inspect smoke evaluation.
- M-6 added multiple task families and deterministic scorers.
- M-7 stress-tested the benchmark itself.
- M-8 packaged the final developer report, artifact index, and roadmap.

Cycles 10-12 therefore had a narrow purpose: determine whether closure remained valid, not expand the benchmark.

Approach

The reporting approach for cycles 10-12 was chronological. Each cycle had three supplied session records: researcher, worker, and auditor. The report uses those records and the supplied closure audit as source material.

No new correctness audit was performed for the underlying prototype. The validation status reported here comes from the supplied audit records, which consistently state:

- pytest: 31 passed.
- Stress runner: `matched_expectations`: 11/11.
- Multi-family Inspect: 8 samples, accuracy 1.000.
- `promise_check`: exit 0 with known non-blocking bookkeeping warnings.
- `org_check`: green.

The report also checked that the final artifacts named in the closure audit are present in the workspace:

- `reports/final/final_report.md`
- `reports/final/artifact_index.md`
- `reports/final/roadmap.md`

Two existing figures remain useful context for understanding the already-validated campaign result. They are not new cycle 10-12 outputs.

Figure artifact: Four-family deterministic verdict matrix for the validated task families.. SVG source: [alignment-test-factory/examples/families/multi_family_verdict_matrix.svg](#).

Figure artifact: Stress probe outcomes showing 11 matched expectations across gaming, ambiguity, false-positive, false-negative, and trace-integrity probes.. SVG source: [alignment-test-factory/examples/stress/benchmark_stress_matrix.svg](#).

Findings

Cycle 10: First Closure Confirmation

Cycle 10 confirmed that no new research question remained open.

The researcher session 73372672-b588-4934-8719-9756a0be3e05 stated that closure was the active sub-topic. It recorded that M-1 through M-8 were independently validated, the final artifacts were present, and no new build, run, or fanout was recommended.

The worker session 26b13e01-66d0-45b2-abe3-2df881d91be0 performed no build and ran no commands. Its result was a closure statement: all planned milestones had been validated, and the final artifacts were `final_report.md`, `artifact_index.md`, and `roadmap.md`.

The auditor session f54a7501-9328-4db4-b264-55313f52dd6b issued a VALIDATED decision. It reported no CRITICAL or MODERATE findings. The only remaining issue was the known promise_check bookkeeping warnings, classified as non-blocking and unrelated to the validated factory artifacts.

The cycle 10 rationale was that the run had converged on a practical trace-centered evaluation pattern: task specs, safe synthetic fixtures, deterministic trace validation, task requirement checks, evidence-linked scoring, Inspect-compatible logs, benchmark stress testing, and final developer documentation.

Cycle 11: Repeated Closure Confirmation

Cycle 11 repeated the same closure result and did not open a new implementation direction.

The researcher session 74482b68-fdc2-46ab-838a-b7bd262262b1 stated that the audit report validated M-1 through M-8 with no unresolved CRITICAL or MODERATE issues. It explicitly recommended no build or run. It also recorded the closure falsification criteria: closure would be ruled out by a missing final artifact, a failing core validation command, unsafe benchmark content, or an overclaim that the synthetic prototype validates arbitrary real agents.

The worker session 965de2e7-a08b-4be0-a047-094e91e88e39 performed no build and ran no commands. It confirmed that this was a closure-only cycle and that future work, if started as a separate campaign, should begin from reports/final/roadmap.md.

The auditor session f7cd88ae-alf2-4c84-bf38-d3dee4865ba4 again issued VALIDATED. Its rationale restated the validated factory mechanism:

high-level safety property -> provider-agnostic TaskSpec -> benign scripted trajectory/runtime trace -> trace integrity validation -> task trace requirement validation -> deterministic evidence-linked scorers -> Inspect-compatible logs and summaries -> stress-tested benchmark report -> developer-facing final documentation.

A TaskSpec is the provider-agnostic task specification: it describes the task, policies, expected trace requirements, labels, predicates, and rubric gates without binding the benchmark to one model provider or agent interface. A trace is the recorded sequence of events from an agent or scripted trajectory. A scorer is the deterministic checker that turns trace evidence into pass, fail, inconclusive, or invalid-trace verdicts.

Cycle 12: Final Closure Confirmation

Cycle 12 closed the campaign without adding work.

The researcher session 5e9a5c52-6a2d-457e-980f-0f0f4a5236c2 stated that the full campaign was validated and that no research, build, or test cycle should be opened. It identified the next work only as future-campaign material: packaging and stable command-line interfaces, real-agent trace ingestion, and first-class scorer evidence bundles.

The worker session 38c13013-387f-4779-bd0b-0d7833254626 performed no build and ran no commands. It confirmed that no new task-family, benchmark, or tooling work was opened.

The auditor session f9ffd180-8215-414a-bdce-9cf5775a6078 issued the final VALIDATED decision for closure. It reported no CRITICAL or MODERATE findings. It also stated that no closure falsification criterion was triggered: no missing final artifact, no failing core validation, no unsafe benchmark content, no overclaim of arbitrary real-agent coverage, and no unresolved CRITICAL or MODERATE audit finding.

The auditor’s cumulative note is the clearest summary of what the campaign achieved: it gives an open-source developer a practical path for testing agentic failures that one-turn jail-break lists miss, including tool overreach, provenance gaps, uncertainty and escalation failures, delegation drift, evaluator gaming attempts, malformed traces, and final-answer/trace mismatches.

Discussion

Cycles 10-12 are best read as closure validation, not as late-stage development. Their significance is that three consecutive researcher-worker-auditor cycles agreed that the campaign should stop rather than expand.

That matters because benchmark projects can drift into adding more examples before the core mechanism is stable. The closure records did not recommend more task-family expansion. They instead pointed future work toward engineering maturity: packaging, stable command-line interfaces, real-agent trace ingestion, and scorer evidence bundles.

The campaign’s final answer to the core question is therefore bounded but affirmative. Within a safe synthetic prototype, it can turn high-level agentic safety properties into executable, reproducible, trace-scored evaluations. It does not claim to validate arbitrary real-world agents. It shows a practical factory pattern that can be inspected, extended, and connected to existing evaluation infrastructure such as Inspect [1].

The known residual issue is bookkeeping-only: `promise_check` still reports warnings about managed-report or manager-assessment records. The supplied audit records classify those warnings as non-blocking because they do not affect the validated artifacts, the prototype’s behavior, or the closure decision.

Open Questions

No open questions remain inside this completed campaign.

Future work belongs in a new campaign or implementation track. The closure records consistently identify three highest-priority next steps:

- Package the prototype as an installable Python module with stable command-line interfaces.
- Add real-agent trace ingestion adapters so external agent runs can be converted into the validated trace format.
- Make scorer evidence bundles first-class so developers can inspect why each deterministic verdict was reached.

The closure records also warn against adding new task families before those engineering tracks are complete.

References

[1] UK AI Security Institute, “Inspect AI: Framework for Large Language Model Evaluations,” 2024. <https://inspect.aisi.org.uk/>

Appendix: Implementation Details

Code Organization

The active implementation remains the validated M-1 through M-8 artifact set. No cycle 10-12 code was added.

The refreshed MANIFEST.md records 60 tracked text files and 11,300 tracked text lines across the current factory workspace. The main implementation areas are:

- alignment-test-factory/src/alignment_test_factory/: schema models, runtime, trace validation, scorers, task-family trajectories, and stress probes.
- alignment-test-factory/tools/: command-line scripts for schema export, spec validation, runtime execution, Inspect smoke runs, multi-family runs, and stress runs.
- alignment-test-factory/evals/: Inspect task adapters for the permission smoke eval and multi-family smoke suite.
- alignment-test-factory/examples/: valid and invalid task specs, generated runtime traces, deterministic summaries, Inspect summaries, Inspect log manifests, and benchmark stress artifacts.
- tests/: six pytest files covering schema validation, toy runtime behavior, Inspect summaries, multi-family scoring, and stress behavior.
- reports/final/: final developer-facing report, artifact index, and roadmap.
- reports/cycles/: periodic reports for cycles 1-3, 4-6, and 7-9.

Test Results Reported by Closure Records

No commands were run in cycles 10-12. The closure cycles relied on the previously audited validation state:

Validation item	Reported result
Pytest suite	31 passed
Benchmark stress runner	matched_expectations: 11/11
Multi-family Inspect smoke	8 samples, accuracy 1.000
promise_check	exit 0 with known non-blocking bookkeeping warnings
org_check	green

Session References

Cycle	Role	Session ID	Content
10	researcher	73372672-b588-4934-8719-9756a0be3e05	Closure topic; M-1 through M-8 validated; no new build or run recommended.
10	worker	26b13e01-66d0-45b2-abe3-2df881d91be0	No build; no commands; final artifacts confirmed.
10	auditor	f54a7501-9328-4db4-b264-55313f52dd6b	VALIDATED; no CRITICAL/MODERATE findings; bookkeeping warnings non-blocking.

Cycle	Role	Session ID	Content
11	researcher	74482b68-fdc2-46ab-838a-b7bd262262b1	Closure assessment; no empirical ladder opened; future work should start from roadmap.
11	worker	965de2e7-a08b-4be0-a047-094e91e88e39	No build; no commands; closure-only cycle.
11	auditor	f7cd88ae-a1f2-4c84-bf38-d3dee4865ba4	VALIDATED; final package and milestones validated.
12	researcher	5e9a5c52-6a2d-457e-980f-0f0f4a5236c2	Final closure; no new task-family or tooling work recommended.
12	worker	38c13013-387f-4779-bd0b-0d7833254626	No build; no commands; final artifacts confirmed.
12	auditor	f9ffd180-8215-414a-bdce-9cf5775a6078	Final VALIDATED closure decision; no remaining research, build, test, or audit action.

Cross-Reference Map

Source	Related artifact	Relationship
Cycle 10-12 audit records	reports/final/final_report.md	Closure records validate the final synthesis as complete.
Cycle 10-12 audit records	reports/final/artifact_index.md	Closure records validate the developer artifact map as present.
Cycle 10-12 audit records	reports/final/roadmap.md	Closure records identify the roadmap as the starting point for future work.
M-7 stress validation	alignment-test-factory/examples/stress/benchmark_stress_results.json	Closure records rely on the previously audited 11/11 matched stress expectations.
M-6 Inspect validation	alignment-test-factory/examples/inspect/multi_family_inspect_score_summary.json	Closure records rely on the previously audited 8-sample multi-family Inspect result.
Reporter manifest refresh	MANIFEST.md	Current snapshot now includes reports/cycles/report_cycles_7-9.md and updated totals.

Record Gaps

There are no missing build logs for cycles 10-12 because no build or test commands were run in those cycles. This is an intentional closure state, not an unexplained absence of evidence.